Exercise Sheet 7

Exercise 1 – Van der Waals interaction of graphene with Au and $\mathrm{SiO_2}$

We are given the van der Waals interaction energy per unit area between a graphene monolayer and a substrate:

$$\frac{E(D)}{A} = \frac{K}{D^9} - \frac{C_3}{D^3}, \quad K > 0,\ C_3 > 0,$$

where the first term is short-range Pauli repulsion and the second is long-range attractive van der Waals interaction.

(a) Equilibrium distance D_0

At equilibrium, $\frac{E(D)}{A}$ is minimized:

$$\frac{d}{dD}\left(\frac{E(D)}{A}\right) = 0.$$

Compute the derivative:

$$\frac{d}{dD}\left(\frac{K}{D^9} - \frac{C_3}{D^3}\right) = -9\frac{K}{D^{10}} + 3\frac{C_3}{D^4}.$$

Set equal to zero:

$$-9\frac{K}{D_0^{10}} + 3\frac{C_3}{D_0^4} = 0 \quad \Rightarrow \quad 3\frac{C_3}{D_0^4} = 9\frac{K}{D_0^{10}}.$$

Multiply both sides by D_0^{10}:

$$3C_3 D_0^6 = 9K \quad \Rightarrow \quad D_0^6 = \frac{3K}{C_3}.$$

Thus,

$$\boxed{D_0 = \left(\frac{3K}{C_3}\right)^{1/6}}.$$

(b) Binding energy per area at equilibrium

Substitute D_0 back into $E(D)/A$. First compute:

$$\frac{1}{D_0^3} = \left(\frac{C_3}{3K}\right)^{1/2}, \quad \frac{1}{D_0^9} = \left(\frac{1}{D_0^3}\right)^3 = \left(\frac{C_3}{3K}\right)^{3/2}.$$

Therefore:

$$\left.\frac{E(D)}{A}\right|_{D=D_0} = K \cdot \left(\frac{C_3}{3K}\right)^{3/2} - C_3 \cdot \left(\frac{C_3}{3K}\right)^{1/2}.$$

Factor out $\left(\frac{C_3}{3K}\right)^{1/2}$:

$$= \left(\frac{C_3}{3K}\right)^{1/2}\left[K \cdot \frac{C_3}{3K} - C_3\right] = \left(\frac{C_3}{3K}\right)^{1/2}\left[\frac{C_3}{3} - C_3\right] = \left(\frac{C_3}{3K}\right)^{1/2}\left(-\frac{2}{3}C_3\right).$$

Simplify:

$$= -\frac{2}{3}C_3 \cdot \sqrt{\frac{C_3}{3K}} = -\frac{2}{3} \cdot \frac{C_3^{3/2}}{(3K)^{1/2}} = -\frac{2}{3\sqrt{3}} \cdot \frac{C_3^{3/2}}{K^{1/2}}.$$

Thus,

$$\boxed{\left.\frac{E(D)}{A}\right|_{D=D_0} = -\frac{2}{3\sqrt{3}} \cdot \frac{C_3^{3/2}}{K^{1/2}}}.$$

(c) Comparison for Au vs. $\mathrm{SiO_2}$

Given: $C_3^{(\mathrm{Au})} = 3C_3^{(\mathrm{SiO_2})}$, and K is identical.

(a) Ratio of equilibrium distances:

$$D_0 \propto C_3^{-1/6} \quad \Rightarrow \quad \frac{D_0^{(\mathrm{Au})}}{D_0^{(\mathrm{SiO_2})}} = \left(\frac{C_3^{(\mathrm{Au})}}{C_3^{(\mathrm{SiO_2})}}\right)^{-1/6} = 3^{-1/6}.$$

Thus,

$$\boxed{\frac{D_0^{(\mathrm{Au})}}{D_0^{(\mathrm{SiO_2})}} = 3^{-1/6} \approx 0.83}.$$

(b) Ratio of binding energy magnitudes: From part (b), $|E_{\min}/A| \propto C_3^{3/2}$, so:

$$\frac{|E_{\min}^{(\mathrm{Au})}/A|}{|E_{\min}^{(\mathrm{SiO_2})}/A|} = \left(\frac{C_3^{(\mathrm{Au})}}{C_3^{(\mathrm{SiO_2})}}\right)^{3/2} = 3^{3/2} = 3\sqrt{3} \approx 5.20.$$

Thus,

$$\boxed{\frac{|E_{\min}^{(\mathrm{Au})}/A|}{|E_{\min}^{(\mathrm{SiO_2})}/A|} = 3\sqrt{3} \approx 5.2}.$$

Qualitative comment: Graphene binds more strongly to Au than to SiO_2: the binding energy is over 5 times larger in magnitude, and the equilibrium distance is $\sim 17\%$ smaller. This implies significantly stronger adhesion to the Au surface. Consequently, exfoliation (e.g., via Scotch-tape method) would be more difficult on Au than on SiO_2. The stronger vdW attraction on Au arises from its higher polarizability compared to the insulating SiO_2.

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Exercise 2 – One-dimensional ionic crystal (alternating $\pm q$)

Consider an infinite chain of alternating charges $\pm q$ with lattice spacing R, and short-range repulsion between nearest neighbors: A/R^n, where $n > 1$.

The total potential energy per ion pair is:

$$U(R) = U_{\mathrm{Coul}}(R) + U_{\mathrm{rep}}(R).$$

(a) Madelung constant α

Take a reference ion with charge $+q$ at position 0. Its Coulomb interaction with all other ions:

$$U_{\mathrm{Coul}} = \frac{1}{4\pi\varepsilon_0}\sum_{m\neq 0}\frac{q_i q_j}{r_{ij}}.$$

For alternating $\pm q$, neighbors at positions $\pm R, \pm 2R, \pm 3R, \ldots$ have charges $-q, +q, -q, \ldots$

So interaction energy of the $+q$ ion at 0:

$$U_{\mathrm{Coul}}^{(+)} = \frac{1}{4\pi\varepsilon_0}\left[\frac{(+q)(-q)}{R} + \frac{(+q)(+q)}{2R} + \frac{(+q)(-q)}{3R} + \frac{(+q)(+q)}{4R} + \cdots\right]$$

$$= -\frac{q^2}{4\pi\varepsilon_0 R}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = -\frac{q^2}{4\pi\varepsilon_0 R}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}.$$

The alternating harmonic series converges to $\ln 2$:

$$\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k} = \ln 2.$$

Thus,

$$U_{\mathrm{Coul}}^{(+)} = -\frac{q^2}{4\pi\varepsilon_0 R}\ln 2.$$

By symmetry, the $-q$ ion has the same interaction energy. So the Coulomb energy per ion pair ($+q$ and $-q$) is:

$$U_{\mathrm{Coul}}^{\mathrm{pair}} = 2\times U_{\mathrm{Coul}}^{(+)} = -\frac{2q^2}{4\pi\varepsilon_0 R}\ln 2 = -\frac{q^2}{2\pi\varepsilon_0 R}\ln 2.$$

By definition, Madelung energy per ion pair for a 1D chain is:

$$U_{\mathrm{Coul}}^{\mathrm{pair}} = -\alpha\frac{q^2}{4\pi\varepsilon_0 R}.$$

Matching:

$$-\alpha\frac{q^2}{4\pi\varepsilon_0 R} = -\frac{q^2}{2\pi\varepsilon_0 R}\ln 2 \quad\Rightarrow\quad \alpha = 2\ln 2.$$

Wait: this seems contradictory to standard convention. Actually, in literature, the Madelung constant for 1D alternating chain is usually defined such that the energy per ion is:

$$U_{\mathrm{Coul}}^{\mathrm{per\ ion}} = -\alpha\frac{q^2}{4\pi\varepsilon_0 R},$$

so from above $U^{(+)}_{\text{Coul}} = -\frac{q^2}{4\pi\varepsilon_0 R}\ln 2$, hence $\alpha = \ln 2$.

Indeed, confirm: For 3D NaCl, Madelung constant is defined via energy per ion pair $= -\alpha\frac{q^2}{4\pi\varepsilon_0 r_0}$, with $\alpha = 1.747$. In 1D, energy per ion pair is:

$$U^{\text{pair}}_{\text{Coul}} = -2\cdot\frac{q^2}{4\pi\varepsilon_0 R}\ln 2 = -\frac{q^2}{4\pi\varepsilon_0 R}(2\ln 2).$$

So if the convention is energy per ion pair $= -\alpha\frac{q^2}{4\pi\varepsilon_0 R}$, then:

$$\boxed{\alpha = 2\ln 2}.$$

However, many textbooks (e.g., Kittel) define Madelung constant via energy per ion, not per pair. The problem says "compute the Madelung constant α of the infinite chain," and part (c) expresses $U(R_0)$ proportional to N, with N ions $= N/2$ pairs.

In part (c), final expression is:

$$U(R_0) = -\ln 2\cdot\frac{Nq^2}{4\pi\varepsilon_0 R_0}\left(1-\frac{1}{n}\right).$$

Since this has $\ln 2$ (not $2\ln 2$), and coefficient $\frac{Nq^2}{4\pi\varepsilon_0 R_0}$, it implies:

$$U_{\text{Coul}} = -\ln 2\cdot\frac{Nq^2}{4\pi\varepsilon_0 R},$$

i.e., energy per ion $= -\ln 2\cdot\frac{q^2}{4\pi\varepsilon_0 R}$, so the Madelung constant (per ion) is $\alpha = \ln 2$.

But to match standard conventions: In 1D, the Madelung constant for the alternating chain is indeed $\alpha = 2\ln 2$ **per ion pair**, or $\alpha = \ln 2$ **per ion**.

Given part (c) result contains $\ln 2$, and multiplies by N (number of ions), the problem uses definition:

$$U_{\text{Coul}} = -\alpha\frac{q^2}{4\pi\varepsilon_0 R}\times(\text{number of ions}).$$

Thus,

$$\boxed{\alpha = \ln 2}.$$

(b) Equilibrium spacing R_0

Total energy for N ions ($N/2$ pairs), including repulsion only between nearest neighbors (each ion has one nearest neighbor of opposite sign; but each repulsive pair counted once):

Repulsive term: nearest-neighbor repulsion $\frac{A}{R^n}$ per pair. With $N/2$ pairs:

$$U_{\text{rep}} = \frac{N}{2}\cdot\frac{A}{R^n}.$$

Coulomb term (from part a):

$$U_{\text{Coul}} = -\alpha\cdot\frac{Nq^2}{4\pi\varepsilon_0 R} = -\ln 2\cdot\frac{Nq^2}{4\pi\varepsilon_0 R}.$$

So total energy:

$$U(R) = -\ln 2\cdot\frac{Nq^2}{4\pi\varepsilon_0 R} + \frac{N}{2}\frac{A}{R^n}.$$

Minimize w.r.t. R:

$$\frac{dU}{dR} = \ln 2\cdot\frac{Nq^2}{4\pi\varepsilon_0 R^2} - \frac{N}{2}\cdot n\frac{A}{R^{n+1}} = 0.$$

Cancel N, rearrange:

$$\ln 2\cdot\frac{q^2}{4\pi\varepsilon_0 R^2} = \frac{nA}{2R^{n+1}}.$$

Multiply both sides by R^{n+1}:

$$\ln 2\cdot\frac{q^2}{4\pi\varepsilon_0}R^{n-1} = \frac{nA}{2}.$$

Solve for $R = R_0$:

$$R_0^{n-1} = \frac{2nA}{\ln 2}\cdot\frac{4\pi\varepsilon_0}{q^2} = \frac{8\pi\varepsilon_0 nA}{q^2\ln 2}.$$

Thus,

$$\boxed{R_0 = \left(\frac{8\pi\varepsilon_0 nA}{q^2\ln 2}\right)^{1/(n-1)}}.$$

Alternatively, express A in terms of R_0 for part (c): From above:

$$A = \frac{\ln 2}{2n}\cdot\frac{q^2}{4\pi\varepsilon_0}R_0^{n-1}.$$

(c) Show:

$$U(R_0) = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}\left(1 - \frac{1}{n}\right).$$

Using $U(R) = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R} + \frac{N}{2}\frac{A}{R^n}$, and substitute A from equilibrium condition.
From derivative condition:

$$\frac{N}{2}n\frac{A}{R_0^{n+1}} = \ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0^2} \quad \Rightarrow \quad \frac{N}{2}\frac{A}{R_0^n} = \frac{1}{n} \cdot \ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}.$$

Thus repulsive energy at R_0 is:

$$U_{\text{rep}}(R_0) = \frac{N}{2}\frac{A}{R_0^n} = \frac{\ln 2}{n} \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}.$$

Coulomb energy:

$$U_{\text{Coul}}(R_0) = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}.$$

Therefore total:

$$U(R_0) = U_{\text{Coul}} + U_{\text{rep}} = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0} + \frac{\ln 2}{n} \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0} = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}\left(1 - \frac{1}{n}\right).$$

$$\boxed{U(R_0) = -\ln 2 \cdot \frac{Nq^2}{4\pi\varepsilon_0 R_0}\left(1 - \frac{1}{n}\right)}.$$

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Exercise 3 – Ionic crystal made of identical atoms

(a) Binding energy of hypothetical $\mathrm{Na^+Na^-}$ crystal

Assume fcc structure (same as NaCl), with nearest-neighbor separation $R = 3.72 \ = 3.72 \times 10^{-10}$ m, equal to metallic Na lattice spacing (note: for fcc, nearest-neighbor distance is $a/\sqrt{2}$, but here they directly give R as atomic separation).

To form $\mathrm{Na^+Na^-}$ from neutral Na atoms, we need:

- Ionize one Na atom: cost = ionization energy $I = +5.14$ eV, - Attach electron to another Na atom: gain = electron affinity $A_{\text{Na}} = -0.78$ eV (note: affinity is negative when energy is released; here value is given as -0.78 eV, so energy change is -0.78 eV).

Net energy to form one $\mathrm{Na^+Na^-}$ pair from 2 Na atoms:

$$E_{\text{pair}}^{\text{form}} = I + A_{\text{Na}} = 5.14 \text{ eV} + (-0.78 \text{ eV}) = +4.36 \text{ eV}.$$

So endothermic: requires 4.36 eV per ion pair.

Then, the Coulomb attraction (Madelung energy) stabilizes the crystal. For fcc, Madelung constant $\alpha = 1.747$. Energy per ion pair:

$$U_{\text{Madelung}} = -\alpha\frac{e^2}{4\pi\varepsilon_0 R}.$$

We compute:

$$\frac{e^2}{4\pi\varepsilon_0} = 1.44 \text{ eV} \cdot \quad \text{(standard constant)}.$$

So:

$$U_{\text{Madelung}} = -1.747 \cdot \frac{1.44 \text{ eV} \cdot}{3.72} = -1.747 \cdot 0.3871 \text{ eV} \approx -0.676 \text{ eV}.$$

Thus total binding energy **per ion pair** (i.e., per 2 atoms):

$$E_{\text{tot}}^{\text{pair}} = (+4.36 \text{ eV}) + (-0.676 \text{ eV}) = +3.684 \text{ eV}.$$

So net **positive**: the crystal is unstable — formation costs energy.

Per atom:

$$E_B^{\mathrm{Na^+Na^-}} = \frac{+3.684}{2} \approx +1.84 \text{ eV/atom}.$$

Compare with cohesive energy of metallic Na: -1.13 eV/atom. Thus, the hypothetical ionic crystal is *much less stable* — in fact, unbound.

$$\boxed{E_{\text{bind}} \approx +1.84 \text{ eV/atom} \quad \text{(unstable)}}$$

(b) Compare with NaCl crystal

For NaCl, formation per Na^+Cl^- pair from neutral atoms:
- Ionize Na: $+5.14$ eV, - Attach electron to Cl: electron affinity $A_{\text{Cl}} = -3.61$ eV,
Net ion-pair formation cost:

$$5.14 \text{ eV} - 3.61 \text{ eV} = +1.53 \text{ eV}.$$

Madelung energy: use nearest-neighbor distance. For NaCl, lattice constant $a = 5.6402$, and in rock-salt (fcc) structure, nearest-neighbor distance is $R = a/2 = 2.8201$.

$$U_{\text{Madelung}}^{\text{NaCl}} = -\alpha \frac{e^2}{4\pi\varepsilon_0 R} = -1.747 \cdot \frac{1.44 \text{ eV} \cdot}{2.8201} = -1.747 \cdot 0.5106 \text{ eV} \approx -0.892 \text{ eV}.$$

Total energy per pair:

$$E_{\text{tot}}^{\text{NaCl}} = +1.53 \text{ eV} - 0.892 \text{ eV} = +0.638 \text{ eV}.$$

Still positive? But we know NaCl is stable. What's missing? Short-range repulsion!

In real ionic crystals, there is a **repulsive term** (Born repulsion) that is essential: the total lattice energy is:

$$U = -\alpha \frac{e^2}{4\pi\varepsilon_0 R} + \frac{B}{R^n},$$

and minimization yields a deeper minimum. Moreover, the lattice energy is usually evaluated at equilibrium R_0, not at arbitrary distance.

But here, the problem asks to *compare* — and we are told to assume both have fcc structure. However, the Madelung term alone is insufficient; but given the data, likely the intention is to use the Madelung energy at the *actual* lattice spacing (i.e., NaCl at its measured a, and NaNa at metallic Na spacing), and include only Madelung + ionization/affinity.

In fact, the experimental lattice energy of NaCl is about -7.9 eV per mole of ion pairs,i.e. -0.082 eV/pair? No wait: per mole,$787 kJ/mol 8.16 eV$ per ion pair. Actually:

$$U_{\text{lattice}}^{\text{NaCl}} \approx -7.9 \text{ eV} \quad \text{(per ion pair)}.$$

Wait: 787 kJ/mol $\div$ 96.485 kJ/mol/eV = 8.16 eV per mole of ion pairs $\rightarrow$ per ion pair, -8.16 eV. Yes.

But our Madelung-only estimate is only -0.89 eV, so clearly missing Born repulsion and proper minimization.

However, within the simplified model given (only Madelung + ionization/affinity), and using the *actual* equilibrium spacing for NaCl and the metallic spacing for NaNa, we can still compare the *Coulomb contribution*.

But problem statement says: "assuming both crystals have the fcc structure." It doesn't say same spacing. So we use: - For Na^+Na^-: use $R = 3.72$ (given), - For Na^+Cl^-: use measured lattice constant $a = 5.6402$, so $R = a/2 = 2.8201$,

and compute total energy as:

$$E_{\text{tot}} = (I + A) - \alpha \frac{e^2}{4\pi\varepsilon_0 R}.$$

We already computed: - Na^+Na^-: $+4.36 - 0.676 = +3.68$ eV/pair, - Na^+Cl^-: $+1.53 - 0.892 = +0.64$ eV/pair.

Both are positive — but NaCl is actually stable. Why? Because the repulsive term is *not* negligible — and more importantly, the equilibrium spacing minimizes total energy, not just Coulomb.

But the question likely expects this calculation *ignoring repulsion*, to illustrate why Na^+Na^- does not form: even before accounting for repulsion, the net "ionic formation + Madelung" cost is much higher for NaNa than for NaCl.

Alternatively, perhaps the Madelung energy should be calculated at the *equilibrium* spacing *predicted* by minimization.

However, the problem gives *no* repulsive parameters for either crystal, so we cannot minimize. Thus, we proceed with the given spacings.

Conclusion:
- For Na^+Na^-: net energy cost $+3.68$eV per pair, - For Na^+Cl^-: net energy cost $+0.64$eV per pair.

Even though both are positive in this model, the cost is over $5\times$ larger for NaNa. In reality, including attractive dispersion and repulsive terms, NaCl becomes stable, while NaNa remains unstable — consistent with the fact that identical-atom ionic crystals don't exist.

Thus, Na^+Na^- crystal is far less stable than NaCl, and also less stable than metallic Na (cohesive energy -1.13 eV/atom $= -2.26$ eV/pair).

$$\boxed{\begin{aligned} E^{\text{pair}}_{\text{Na}^+\text{Na}^-} &\approx +3.68 \text{ eV}, \\ E^{\text{pair}}_{\text{Na}^+\text{Cl}^-} &\approx +0.64 \text{ eV} \quad \text{(simplified model)}. \end{aligned}}$$

The smaller ionization–affinity gap for NaCl (1.53 eV vs. 4.36 eV) and shorter bond length make the Madelung stabilization more effective, explaining why heteronuclear ionic crystals form but homonuclear ones do not.